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EFY Note

The source code of this project is included in this month's EFY DVD and is also available for free download on www.efymag.com website.

transistor T5 should be as per the coil current rating of the relay. Transistor with higher current capability should be used for higher coil current rating of the relay.

Software program

The program for the microcontroller is written in 'C' and compiled using AVR Studio 4. The hex code generated can be programmed into the microcontroller using any standard programmer board. In EFY Lab, we programmed it using EFY-KnS AVR programmer board. AVR Studio 4 can be downloaded free of cost from the Atmel website. The guide to use the software is also available on this website.

Construction and testing

Actual-size, single-side PCB for the smart power control transmitter circuit is shown in Fig. 4 and its component layout in Fig. 5. PCB layout of receiver circuit is shown in Fig. 6 and its component layout in Fig. 7. Assemble the components on the PCBs. Use of IC bases is recommended for all the ICs on the PCBs.

Install IRTX1 and IRRX1 on the opposite door frames at the entry point such that the transmitter directly faces the receiver. Similarly, IRTX2 and IRRX2 should be installed on the opposite door frames at the exit point.

RL1 and RL2 are not PCB mounted relays; they are to be connected externally.

When someone enters through the entry gate, you should see the counter increment by '1.' The counter can count up to '99.' When the count reaches '99,' the user has to reset the counter so that it restarts from '0.' When someone leaves the building through the exit gate, the counter decrements by one count. When the count reaches '0,' which means nobody is inside the

PARTS LIST

Semiconductors:

IC1	- LM324 op-amp
IC2	- ATmega32 microcontroller
IC3	- HT12E encoder IC
IC4, IC6	- 7805, 5V regulator
IC5	- HT12D decoder IC
IC7	- 7812, 12V regulator
IC8	- 7555 timer
T1-T4	- BC547 npn transistor
T5	- SL100 npn transistor
T6	- BC557 pnp transistor
LED1-LED3	- 5mm light-emitting diode
D1, D2	- 1N4007 rectifier diode
LDR1	- Light-dependent resistor
IRTX1, IRTX2	- IR LED transmitter
IRRX1, IRRX2	- IR LED receiver
DIS1, DIS2	- LT542 common-anode 7-segment display
RF TX1	- 433MHz RF transmitter
RF RX1	- 433MHz RF receiver

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

R1, R12, R13,	
R15	- 680-ohm
R20, R21, R23,	
R10	- 10-kilo-ohm
R3, R4, R5, R7,	
R8, R9, R16, R19	- 1-kilo-ohm
R2, R6, R11	- 470-ohm
R14	- 1-mega-ohm
R17	- 3.9-kilo-ohm
R18	- 47-kilo-ohm
VR1, VR2	- 20-kilo-ohm preset
VR3	- 4.7-kilo-ohm preset
VR4	- 2.2-mega-ohm preset

Capacitors:

C1, C9	- 470 μ F, 35V electrolytic
C2, C4, C5, C6	- 0.1 μ F ceramic
C7	- 330 μ F, 35V electrolytic
C3	- 10 μ F, 16V electrolytic

Miscellaneous:

Jack	- 9V DC, 250mA power supply (adaptor)
	- 15V DC, 1A power supply (adaptor)
ANT.1, ANT.2	- 10cm hook-up wire antenna
RL1	- 12V, 1C/O relay
RL2	- 12V, 3C/O relay

building, the control system switches off the mains power supply. Thus it saves energy and money.

If you encounter any problem while testing the circuit, first of all check whether power supplies are proper in the circuit. If power supplies are proper but not the display, press reset switch S1 momentarily and observe the display. If the problem still persists, check the various test points as listed in the table to troubleshoot the circuit. ●

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